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title: Burak Dursun - Software Engineer Portfolio

Burak Dursun

Software Engineer | Backend & AI/ML Enthusiast

GitHub | LinkedIn | Medium | Portfolio Website

1. About Me / Profile

A passionate software engineer with a strong foundation in modern web technologies, software architecture, and artificial intelligence. Adept at designing scalable backend systems (RESTful APIs, Clean Architecture), developing performance-critical applications (Multi-threading, Golang, Node.js), and implementing machine learning models for computer vision and data processing. Experienced in leading tech communities and committed to continuous learning, delivering robust solutions, and exploring cloud and DevOps practices. Eager to bring my engineering mindset, problem-solving skills, and technical depth to the innovative teams at Baykar Tech.

2. Technical Skills

- **Programming Languages:** Python, JavaScript/TypeScript, Golang, HTML5/CSS3
- **Backend & Architecture:** Node.js, RESTful APIs, Clean Architecture, Concurrent Programming (Multi-threading), System Design
- **Frontend Technologies:** React.js, Vue.js, Vite, Framer Motion, Tailwind CSS
- **AI & Data Science:** Deep Learning, Machine Learning, CNNs, LSTM, MediaPipe, Data Processing, Web Scraping
- **Databases & DevOps:** SQL / NoSQL Databases, Cloud Concepts, Git/GitHub Version Control

3. Featured Projects

Deep Learning-Based Sign Language Recognition System

- **Overview:** Engineered an AI-driven computer vision system that automatically recognizes Turkish Sign Language (TİD) from video streams.
- **Technical Contribution:** Analyzed human body movements via camera streams using MediaPipe Hands and trained LSTM (Long Short-Term Memory) neural networks to classify temporal sequence data with high accuracy.

- **Technologies:** Python, Deep Learning, LSTM, MediaPipe, OpenCV
- **Link:** github.com/burak-dursunn/Derin-renme-Tabanl-T-rk-aret-Dili-Tan-ma-Sistemi-

Node.js Clean Architecture API

- **Overview:** A comprehensive backend infrastructure developed for an administrative panel, strictly adhering to Clean Architecture principles.
- **Technical Contribution:** Designed a robust RESTful API with separation of concerns. Implemented advanced database operations, dependency injection, and scalable industry best practices for highly maintainable codebases.
- **Technologies:** Node.js, Express, Database Management, Clean Architecture Principles
- **Link:** github.com/burak-dursunn/node-clean-architecture

High-Performance Multi-Threaded Downloader

- **Overview:** A production-grade file downloader designed for maximum network utilization and speed.
- **Technical Contribution:** Leveraged concurrent programming concepts in Python to split large files into chunks and download them simultaneously across multiple threads, significantly reducing execution time.
- **Technologies:** Python, Multi-threading, Network Programming, Concurrency
- **Link:** github.com/burak-dursunn/multi-thread-file-downloader

PemaHome Full-Stack Platform

- **Overview:** A scalable, modern e-commerce platform encompassing both robust backend operations and a dynamic user interface.
- **Technical Contribution:** Utilized Golang for a high-performance backend, ensuring fast request handling and API responses, while delivering a responsive client-side experience using Vue.js.
- **Technologies:** Golang, Vue.js, REST API
- **Link:** github.com/burak-dursunn/pema-home-website

BilgecAI Platform

- **Overview:** An integrated web platform combining front-end interfaces with Python-based AI models.
- **Technical Contribution:** Bridged the gap between machine learning scripts and end-users by designing an interactive Vue.js interface communicating with a Python REST API backend.
- **Technologies:** Vue.js, Python, REST API, Artificial Intelligence
- **Link:** github.com/burak-dursunn/BilgecAi

4. Technical Articles & Publications

I actively research and write engineering articles on Medium, focusing on Artificial Intelligence, System Security, and Technology Trends.

- **How Do Machines Learn to See? A Close Look at CNNs** *(Makinalar Görmeyi Nasıl Öğrenir? CNN'e Yakından Bakış)* Explored the fundamental mathematics and engineering behind Convolutional Neural Networks (CNNs), image processing, and deep learning architectures.
- **The Power of Lava Lamps in Internet Security** *(Lav Lambalarının İnternet Güvenliğindeki Gücü)* Detailed Cloudflare's cryptographic methodologies, discussing random number generation, encryption algorithms, and modern cybersecurity mechanisms.
- **DeepSeek: Low Cost, Big Impact** Analyzed the optimization, cost-efficiency, and disruptive impact of modern Large Language Models (LLMs) on the AI industry.

5. Leadership & Community Experience

Medium Committee Leader | Huawei Student Developers (HSD)

- Managed the editorial processes and content strategy for the HSD Medium engineering blog.
- Monitored writer performances, curated technical topics, and ensured the team successfully met monthly publication targets.
- Led project groups by fostering a culture of knowledge-sharing, mentoring peers on technical writing, and coordinating collaborative software research.